

ICT MID SEM NOTES

Robotics:

- ✓ Automated car assembly line is example of Industrial Automation.
- ✓ Digital computing is 3rd revolution whereas fusion of digital and biological world is 4th
- ✓ 1st industrial rev. 80 yrs 2nd 44 yrs 3rd 1970-2025+ and 4th 2010-2025+
- ✓ Real time visibility, remote monitoring, automation support values IoT
- ✓ IoT needs CPS for feedback-action.
- ✓ **Digital twin** is virtual replica of real world physical system. NOT a static model. Enhance performance and reduce cost.
- ✓ **Data** collect+save—analyze—simulate—evaluate—adjust settings—use product!
- ✓ physical asset +data=digital twin and DT+info=physical asset
- ✓ connects via sensors+IoT data. What connects yes DT!
- ✓ Its origin is NASA Apollo mission. Twins is most transformative tool for Industry 4.0
- ✓ Smart homes are example of CPS
- ✓ Cyber part integrated with physical systems is CPS

- ✓ Physical components include energy sources, transmission and distribution lines, loads, and control devices.
- ✓ Cyber components include EMS, supervisory control and data acquisition (SCADA) systems.
- ✓ CPS—>IIoT, robotics, OT(op.tech), ICS etc.
- ✓ CPS adds real time control and decision making. - **IoT is a subset of CPS** because it focuses on connectivity and data exchange among physical devices, but CPS goes further by emphasizing control, computation, and integration with physical processes.
- ✓ IoT thermostat – data && CPS thermostat—actively adjusts heating cooling.
- ✓ Quantum computing supports CPS and ML helps in sensing attacks on CPS!
- ✓ Cryptography and Blockchain secure IoT and CPS networks. Tamper-proof communication refers to systems or methods designed to prevent unauthorized access, alteration, or interception of messages during transmission. **Ex. Tamper-proof communication in smart grids.**
- ✓ Data mining is the process of discovering patterns, trends, and useful information from large sets of data. It's like finding hidden treasures in a mountain of numbers and text. NLP (Natural Language Processing) is a branch of artificial intelligence that helps

computers understand, interpret, and respond to human language—both written and spoken.

- ✓ Data mining, NLP, AI, Deep learning use large data streams in CPS for predictive maintenance and autonomous system
- ✓ CPS systems required structured development processes benefiting from project management practices.
- ✓ **SoS** is composition of systems collaborate to achieve common objectives.
- ✓ **Operational and Managerial** Independence are key features.
- ✓ Sign of Smart cities: traffic lights reduce congestion, grids manage storage, DT of entire city as Singapore and integrated services.
- ✓ Heterogeneous, emergent, large in scale---SoS
- ✓ Challenges maybe C&C(control and coordination), fault design and new architectures.
- ✓ **Sustainability** offers resilient planning lower emission and Renewable integration.
- ✓ **Robotics** helping in shaping the future, problem solving in 3D, cross disciplinary, career growth, and hands on exp.
- ✓ The word 'robot' was introduced by Czech writer Karel Čapek in his 1920 play R.U.R. (Rossum's Universal Robots).

- ✓ 'robota,' meaning 'forced labor' or 'drudgery', 'statute labor' or 'executive labor'
- ✓ RUR tells story that mass produces artificial workers are called robots.
- ✓ Initially robots were for manual labor but over time they become intelligent.
- ✓ “A well-constructed machine operates on its own; similarly, an automaton (mechanical man) should be made to function everywhere efficiently.” Said in Sanskrutam.
- ✓ Flying machines, automated devices and mechanical birds Samarāṅgaṇa Sutrādhāra in King Bhoja dynasty(11CE)
- ✓ Robots made of metal and wood, animated by mercury and steam.
- ✓ **Yoga Vasistha** (10CE) mentions yantras (machines) that could be programmed to carry out tasks, hinting at ideas of mechanical automation.
- ✓ **Brihat Samhita** (6CE) Mathematics and Engineering – Varahamihira’s work includes references to mechanical contraptions, automated temple doors, and devices to predict celestial movements.
- ✓ **Rigveda** mentions flying chariots and mechanical birds.
- ✓ Karna’s Armor and Arjuna’s Chariots – The Mahabharata describes divine weapons and self-

navigating chariots that could change directions autonomously during battle.

- ✓ Pushpaka Vimanas and Rakshasas are some more from Ramayana.
- ✓ Study of those machines that can replace human beings in regard to physical activity and decision making is *Robotics*.
- ✓ Something that would mimic humans-Scarecrows, paintings, movies, dolls etc. are cultural roots of Robotics.
- ✓ Asimov's 3 Laws of robotics laws were designed to regulate the behavior of robots and ensure their coexistence with humans.
- ✓ 1. A robot may not injure a human being, or, through inaction, allow a human being to come to harm.
- ✓ 2. A robot must obey the orders given to it by human beings except where such orders would conflict with the First Law.
- ✓ 3. A robot must protect its own existence as long as such protection does not conflict with the First or Second Law.
- ✓ Asimov introduced the Zeroth Law: A robot may not harm humanity, or, by inaction, allow humanity to come to harm.(**main law for humanity**)
- ✓ Ethical Issues : Accountability,Bias & fairness,Surveillance,Human-robo interaction,Job displacement.

- ✓ MuJoCo, ROS, Matlab+Simulink, Python libraries are software packages.
- ✓ ROS designed to facilitate robotics applications.
- ✓ ROS is not an operating system but rather a middleware that provides essential tools, libraries, and conventions for building and managing robot software across different platforms.
- ✓ **ROS**-Nodes, Topics, Services, Message, Launch files.
- ✓ Mechanical Components-
Frame(shape), actuators(energy—
>move), joints, wheels for locomotion, End effectors(grippers).
- ✓ Control systems-
microcontrollers(Arduino), embedded systems, FPGAs(hardware), PLCs(industrial control systems)
- ✓ Sensors- proximity(detect nearby), vision systems(LIDAR), gyroscopes(detect orientation), encoders(rotational) and temp. sensors.
- ✓ Power systems-batteries(li-ion), PSUs, fuel cells(alternate source) and capacitors.
- ✓ Actuation-DC/AC motors, stepper-servo motors(small-precise), Pneumatic and hydraulic actuators.(compressed air-fluid pressure).
- ✓ Communication systems-Wifi, Bluetooth, CAN bus, serial communication, Zigbee(wireless data transfer)

- ✓ Operating Systems (OS): ROS (Robot Operating System), VxWorks. Control Algorithms: PID, LQR. Simulators: Gazebo, PyBullet, Webots for testing designs virtually.
- ✓ Mobility-LIDAR(light), IMU, GPS Modules(outdoor), SLAM(auto)
- ✓ **Safety**-Emergency stop, limit switches, surge protectors.
- ✓ **Networking**-Simulators: Gazebo, PyBullet, Webots for testing designs virtually. **Simulators**: Gazebo, PyBullet, Webots for testing designs virtually.
- ✓ AI and ML algorithms allow robots to learn from data, recognize patterns, and make decisions. Ex-humanoid robots with NLP.
- ✓ Jobs-Robotics software engineer, Embedded systems engineer, robotics data scientist, AI/ML eng. For robots.
- ✓ **Upcoming fields**-swarm robotics(multiple coordination), humanoid robotics, medical robotics, autonomous vehicles(robots for self driving cars).
- ✓ **Autonomous system is** system capable of operating in real world environment without any form of external control for period of time.
- ✓ **0 No Autonomy** Human does everything.
- ✓ **1 Operational Assistance** System assists but doesn't act.

- ✓ **2 Partial Automation System** performs some actions.
- ✓ **3 Conditional Operation** Car parks autonomously in known conditions.
- ✓ **4 High Automation**-Car parks itself in most scenarios.
- ✓ **5 Full and Trusted Autonomy**-Car handles everything, everywhere.
- ✓ ROBOT performs first laproscopic(abdomen) surgery without human help.
- ✓ 50% chose randomization and 45.8% chose to harm one and 1.4% to harm two!